

Aerothermal and TPS Analysis of ARRIVAL, an Aerocapture Demonstration at Earth

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Aerocapture is an orbital insertion maneuver accomplished with a single-pass through a destination planet's atmosphere, making use of the aerodynamic forces imparted on the vehicle to modulate the spacecraft's apoapsis from a hyperbolic transfer orbit. Even though this maneuver necessitates an entry vehicle aeroshell with a thermal protection system to safeguard the spacecraft from entry heating, there are significant time and fuel mass savings for destinations like Uranus, Neptune, and Venus. Despite these anticipated savings, this technology has not been demonstrated, making mission planners reluctant to incorporate the technology and mission architecture. This paper describes the completed aerothermal environment predictions and thermal protection system design for ARRIVAL, a proposed Earth technology demonstration intended to reduce aerocapture adoption risk in future NASA missions.

I. Nomenclature

k	=	Boltzmann's constant, J/K
m	=	molecular mass
p	=	pressure, Pa
T	=	temperature, K
t	=	thickness, cm
q	=	heat flux, W/cm^2
ϵ	=	emissivity
λ	=	mean free path, m
μ	=	dynamic viscosity, $Pa \cdot s$
ρ	=	density, kg/m^3
σ	=	standard deviation

subscripts

conv	=	convective
rad	=	radiative
w	=	wall

II. Introduction

In 2022, the National Academy of Sciences (NAS) designated the Ice Giants, Uranus and Neptune, as the top priority for planetary science in the next decade because of their distinctive structures and their critical role in the formation and evolution of the solar system [1]. Work by numerous groups has outlined the feasibility for concept missions to deliver

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scientific payload both to Neptune's moon Triton and Uranus [2, 3]. The Uranus Orbiter and Probe (UOP) mission selected by the NAS is a fully-propulsive vehicle delivery to Uranus with an estimated 13 year transit and nearly 50% mass devoted to the necessary fuel for orbital insertion [1]. The large amount of fuel required for this mission severely constrains the available payload for scientific instruments, thereby limiting the quantity of gathered data. Aerocapture may drastically reduce both the time and fuel required for orbital insertion by making use of the planet's atmosphere to rapidly slow down once at the destination rather than the prolonged deceleration of retro-propulsion. Aerocapture may not only reduce the time to orbit by years and provide mass savings, but it would also open launch opportunities that would otherwise be unavailable [4].

The decades of aerocapture studies have employed different spacecraft control schemes for the atmospheric maneuver, including drag modulation. Atmospheric spacecraft control in many NASA flight programs has been achieved through Bank Angle Modulation (BAM) including Apollo, Mars Science Laboratory (MSL), Mars2020, and Orion. A recent study by Dutta et al. [5] evaluated the feasibility of using a heritage aeroshell design from Mars2020 with BAM aerodynamic control to deliver the conceptual payload from the UOP mission selected by NAS. While the vehicle traverses the atmosphere at hypersonic conditions, an entry aeroshell protects the orbiter with the enclosed science equipment. Accurate predictions of the aeroshell surface conditions—including heating, pressure, and shear stress—are critical to the proper design of a thermal protection system (TPS) which in turn is required for aerocapture and mission success. Work done by the authors in Ref. [6] and Ref. [7] documents the aeroshell concept, aerothermal analysis, thermal protection system (TPS) implementation and sizing, and forward work needed ensure low residual risk for a Uranus Orbiter. Even with the studied effort finding significant savings in time to destination and mass required, there are still a number of real and perceived risks. A report authored by NASA's Science Mission Directorate (SMD) team in 2023 [8] identified potential risks associated with aerocapture when applied to future planetary missions. The highest perceived risks to a mission were due to implementation cost and atmospheric uncertainties that could lead to poor predictions of the aeroheating environments.

The Aerocapture to Reliably Reduce Inertial Velocity with Aerodynamics and Lift (ARRIVAL) project plans to demonstrate aerocapture at Earth using lift modulation by bank angle control in conjunction with a Fully Numeric Predictor-corrector for Aerocapture Guidance (FNPAG) algorithm to solve the bank angle command during flight. Successfully conducting aerocapture at Earth will address many concerns and is the purpose behind ARRIVAL's Earth demonstration. Currently, the project is in a pre-phase A stage with predictions and modeling in place for the vehicle design, instrumentation, guidance, navigation, and control, and aeroheating with thermal protection system sizing.

This paper will briefly introduce aerocapture, lay out the proposed Earth demonstration mission, and designed aeroshell in Section III. Section IV contains analysis of the predicted aeroheating environments including the estimated aeroshell surface conditions and several small vehicle trade studies. The forebody TPS and trade-space for the aftbody heatshields are then outlined in Section V along with a substructure sensitivity study. Finally, conclusions and forward work for the study are summarized in Section VI.

III. Background

A. Aerocapture concept

Aerocapture is an atmospheric maneuver that uses the drag generated during atmospheric traversal to decelerate a vehicle in order to capture a spacecraft into an elliptical orbit from an interplanetary trajectory. To accomplish this maneuver, the cruise stage is jettisoned prior to atmospheric entry interface, and the aerocapture vehicle, made up of the aeroshell and the payload, passes through the gaseous environment. As the vehicle performs the maneuver, interaction with the fluid imparts forces on the vehicle, most importantly drag, resulting in dissipated energy. The vehicle's attitude is modulated within an entry corridor such that atmospheric exit is ensured. The entry corridor is bounded by two trajectories, one where the lift vector is pointed away from the planet and the other with the lift vector pointed toward the planet during the entire operation. These bounding trajectories are referred to as "lift-up" and "lift-down" and aerocapture analysis is characterized through evaluation of these two trajectories. In an actual mission, the vehicle is likely to fly somewhere between the bounding trajectories, the entry corridor. The key differences between the two trajectories is the time spent in and the depth of atmosphere penetrated. These parameters impact the aerothermal environment predictions and contribute to the energy absorbed by the TPS. After atmospheric exit, the aeroshell is jettisoned and the orbiter is deployed. The final two maneuvers are at apoapsis to raise the subsequent periapsis out of the atmosphere, and then at the next periapsis to clean up any residual apoapsis error. The total process is conducted over one period of the captured orbit.

B. Earth Technology Demonstration Con-Ops

At this pre-phase A stage of the project, ARRIVAL evaluated two different mission concepts for complexity and stressing the technology for demonstration. In both concepts, the spacecraft is ejected from an Evolved Expendable Launch Vehicle (EELV) Secondary Payload Adapter (ESPA) payload adapter. The first concept investigated was a Geostationary Transfer Orbit (GTO), shown in Fig. 1, which starts after a perigee lowering maneuver during the geostationary transfer orbit. After atmospheric exit, the aeroshell is jettisoned and a periapsis raise maneuver deploys a small sat into a 500 km science orbit. Figure 2 shows the second concept assessed: a higher energy trajectory which returns a payload from Lunar Return Orbit (LRO) to the same science orbit. The spacecraft and an associated cruise stage exits a Near Rectilinear Halo Orbit (NRHO) after a departure burn and transits to Earth. Prior to atmospheric entry, the aeroshell is released from the cruise stage to start the aerocapture pass similar to the GTO concept. The key difference between the two trajectories is the energy dissipated; 2.4 km/s for the GTO and 3.2 km/s for the LRO trajectory. The ARRIVAL team traded the subsystem impacts from these two options and compared to known relevant application missions such as a Flagship Uranus aerocapture maneuver. This paper will focus on the aerothermal and TPS implications for both concepts.

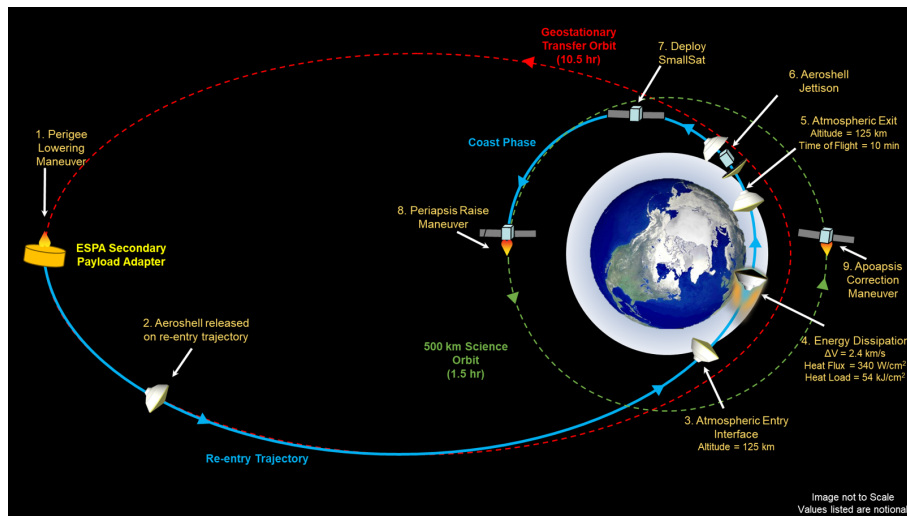


Fig. 1 Geostationary Transfer Orbit Concept.

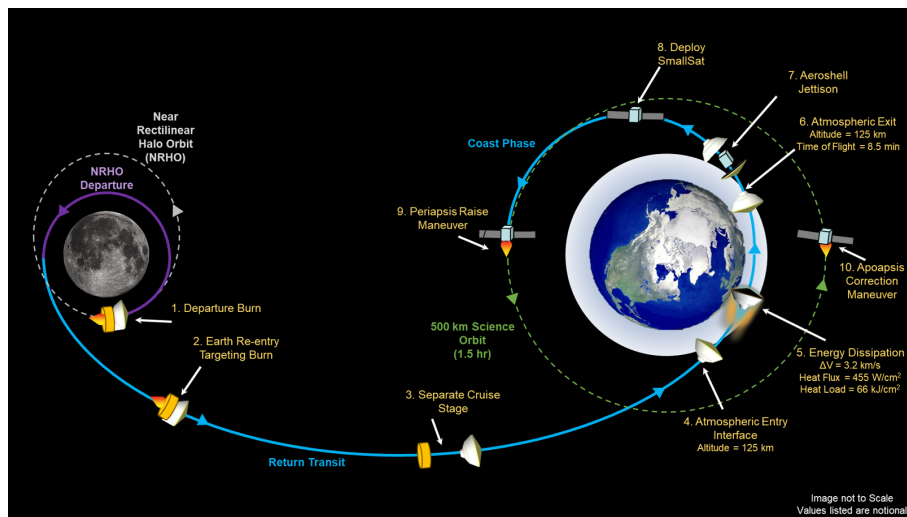


Fig. 2 Lunar Return Orbit Concept.

C. Vehicle Definition

The vehicle design represented in Fig. 3 impacts analysis and choices in each subsystem. However, the vehicle design itself has key requirements from holding a payload satellite to itself fitting within a secondary payload adapter on a launch vehicle. The vehicle must also support re-entry loads, heating, communication and other functions while supporting science instrumentation.

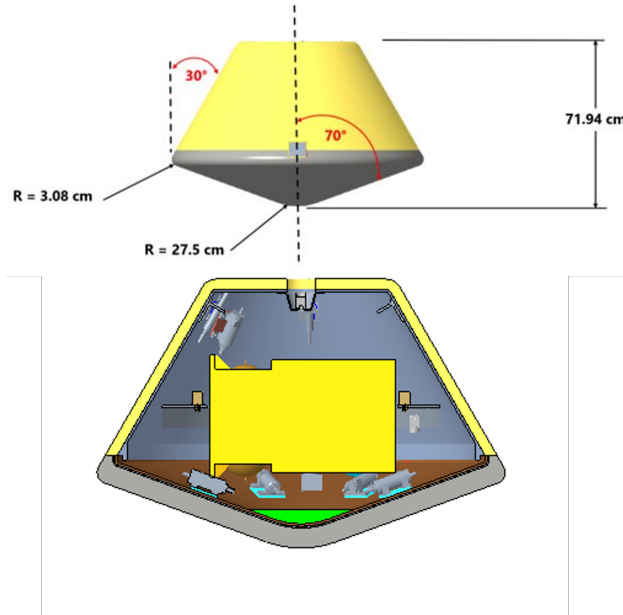


Fig. 3 ARRIVAL Vehicle Design.

The chosen design secondary payload adapters were the ESPA Grande and SpaceX's Falcon adapters which constrained the vehicle mass and volume. The limiting diameter dimension was 46 inches (1.17 m) which encapsulates the aeroshell's 1.1-meter diameter. The aeroshell is a traditional 70-degree sphere-cone shape that is a known aerodynamic quantity and has heritage in many planetary missions. Considering mass growth allowance and pre Authority To Proceed (ATP) mass property rules of standards, AIAA-S120A[9], the maximum permissible value (MPV) of the vehicle and payload was set at 255 kg. With the baselined vehicle shape and mass, re-entry analysis proceeded for each design area including trajectory, aerosciences, and TPS design.

IV. Aeroheating

In this pre-Phase A study, the primary goal of the aerothermodynamics analysis was to roughly bound the aeroheating environments for early vehicle concepts in order to inform the design of the thermal protection system and entry system instrumentation. Early on, the team narrowed down the trajectory options into two concepts of operations, discussed above: a GTO option, and a lunar return option. The aerothermal analysis for both options are presented here. Due to the early nature of the vehicle design process, the focus of the analysis was to build a medium fidelity aerothermal database for use in downstream analyses that spanned a potentially large aerocapture trajectory space and perform targeted studies to support on-going vehicle design activities. In the following subsections, we present the aerocapture trajectory space that was considered for the ARRIVAL vehicle, the computational tools used to estimate aeroheating environments and construct the aerothermal database, and brief summaries of some targeted studies that were performed. A summary of the unmarginated environments is also presented.

A. Trajectory Space

Figure 4 details the trajectory space analyzed in this work. The red and blue trajectory curves represent the lift-up and lift-down trajectories for the lunar return and GTO options, respectively. The lunar return trajectory enters Earth's atmosphere with an atmosphere-relative velocity of 10.8 km/s, while the GTO option is slower, with an entry velocity of about 9.9 km/s. Both options target an exit velocity approaching 7.5 km/s in order to target the desired orbit

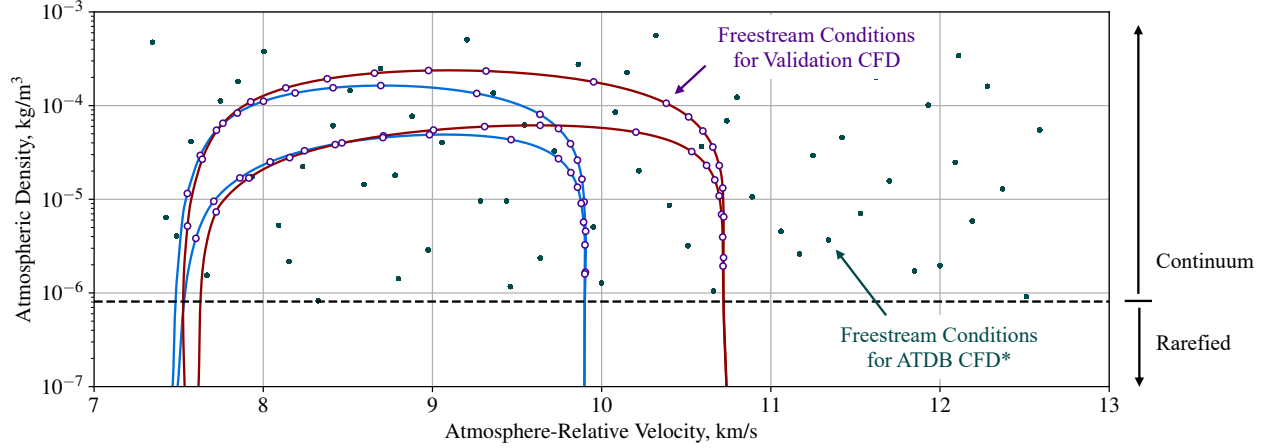


Fig. 4 ARRIVAL trajectory space, including Lunar Return (red) and GTO (blue) concepts and CFD freestream locations.

post-aerocapture. The trajectories shown here are based on three degree-of-freedom trajectory simulations, constrained by an entry inertial velocity, desired exit velocity, and flight control model. The lift-up and lift-down trajectories represent bounding trajectories for each entry and exit condition by imposing full lift-up or lift-down control using a fixed lift-to-drag ratio of 0.25, corresponding to a constant total angle of attack of 17° . For more details on the trajectory analysis, we refer you to a companion paper on the subject [10].

Beyond the atmospheric density and velocity, the NASA Earth Global Reference Atmospheric Model (Earth-GRAM) [11] was used to provide species mole fractions, temperature, and pressure as a function of the geodetic altitude. All remaining freestream quantities were determined with the Mutation++ library [12], by solving for chemical equilibrium at constant temperature and pressure with the given elemental mole fractions. The effective mean free path formulation from Vincenti and Kruger [13] was used in the determination of the freestream Knudsen number $Kn = \lambda/D$, where

$$\lambda = \frac{\mu}{\rho} \sqrt{\frac{\pi m}{2kT}} = \mu \sqrt{\frac{\pi}{2p\rho}}, \quad (1)$$

and μ is the freestream dynamic viscosity, computed using a Chapman-Enskog perturbative solution of the Boltzmann equation for multicomponent gases [12].

B. Computational Analysis Tools

Aerothermal environments were predicted using the NASA LAURA code [14]. LAURA is a 2nd order finite volume Navier-Stokes solver which is commonly used to analyze atmospheric entry vehicles [15]. An 11 species thermochemical nonequilibrium gas model consisting of the following species was considered: N_2 , O_2 , N , O , NO , N_2^+ , O_2^+ , N^+ , O^+ , NO^+ , and e^- . A laminar, non-ablating, fully catalytic wall in radiative equilibrium was assumed for the aerothermal database. Surface radiative heating and flowfield coupling are predicted using the HARA radiation code [16]. During a typical solve, a tangent-slab transport model is used to compute the coupling source term and radiative heat flux at the vehicle surface. For wake radiation the tangent-slab model is sufficient for coupling purposes but inaccurate for surface radiative heat flux [16]. Instead, a ray-tracing transport model is used to compute accurate surface heating for the vehicle aftbody [17]. Additionally, preliminary parametric uncertainty analysis was performed using the LAURA UQ module, which utilizes a non-intrusive polynomial chaos expansion surrogate model with which input parameter uncertainties can be propagated through the CFD model [18]. For the present study, two computational grids were used in general. The first was a forebody-only grid consisting of 24 blocks with $12 \times 12 \times 64$ cells per block. The second grid, shown in Fig. 5, includes the vehicle wake and consists of 48 blocks with $10 \times 10 \times 128$ cells per block. Both computational grids assume lateral symmetry across the vehicle centerline to reduce computational cost. All grids were shock-aligned using LAURA for each flight condition considered.

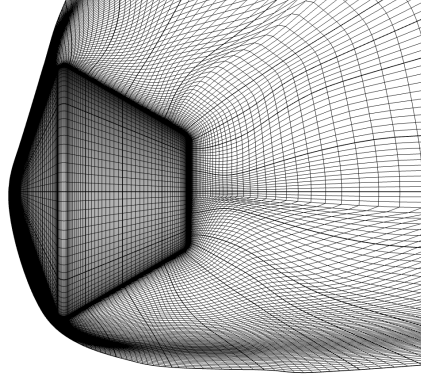


Fig. 5 Example computational grid including the vehicle wake.

C. Aerothermal Database Development

An Aerothermal Database (ATDB) is a surrogate model for CFD simulations that can provide quick estimates of aeroheating quantities of interest across the continuous trajectory space. Such databases are invaluable, especially early in a design, when different trajectory options need to be considered quickly. They also provide a convenient means for estimating the integrated heat load over an entire trajectory, whereas CFD simulations are only valid at discrete points in time. Additionally, ATDBs are routinely used for Monte-Carlo trajectory analyses to study the impact of uncertainties in the atmospheric model and control systems. In this work, we constructed an ATDB using the Bayesian Sutton-Graves approach outlined in [6]. Briefly, this approach utilizes a random scattering of freestream conditions across the desired trajectory space to define a set of CFD cases, which are then solved to provide high fidelity estimates of aeroheating quantities of interest at those freestream conditions, forming a small dataset of solutions. These solutions are then used to infer probability distributions on a small set of parameters governing a surrogate model for the quantities of interest, in this case, the generalized Sutton-Graves model. Once determined, the parametric uncertainties are used to propagate uncertainties through the surrogate model to provide bounded estimates on the quantities of interest.

In this work, we used a Sobel sequence of random points in the freestream density-velocity space to generate a space-filling design of CFD conditions to train the ATDB model. These scattered points are shown throughout the density and velocity space in Fig. 4. Note that the velocity space was extended out to about 12.8 km/s to support alternative trajectory options not presented here. In addition to the space-filling points, we also generated conditions along the lift-up and lift-down trajectories at estimated fractions of the the peak heating conditions. Those points are shown along the trajectory in Fig. 4 and were used to provide more confidence in the estimated environments for those trajectories, as well as serve as a validation set for the ATDB, which was not trained explicitly on those points.

The ATDB itself comprised of a set of Bayesian Sutton-Graves models for approximately 100 body points along the forebody pitching plane of the vehicle. At each body point, a separate model was trained for each quantity of interest, which included convective and radiative heat fluxes, pressure, and shear stress. Additional quantities of interest were derived from these base models. For example, the surface temperature was derived from the radiative equilibrium assumption used in the CFD calculations. Specifically,

$$T_w = \left(\frac{q_{\text{conv}} + q_{\text{rad}}}{\sigma \epsilon} \right)^{\frac{1}{4}}. \quad (2)$$

Once trained, the ATDB provides all quantities of interest at the selected body points as a function of freestream density and velocity. Both mean and 95 % confidence intervals are provided, however only the upper bound of the interval is considered here.

D. Aeroheating Environments Summary

This subsection briefly summarizes the predicted aeroheating environments for the GTO and lunar return trajectory options. All results shown here are unmargined, however margined environments were used for downstream analyses. Figure 6 shows the predicted convective heat flux for each reference trajectory at the vehicle location corresponding to the peak heating location. Both the ATDB predictions and the validation CFD results are shown. In general, the

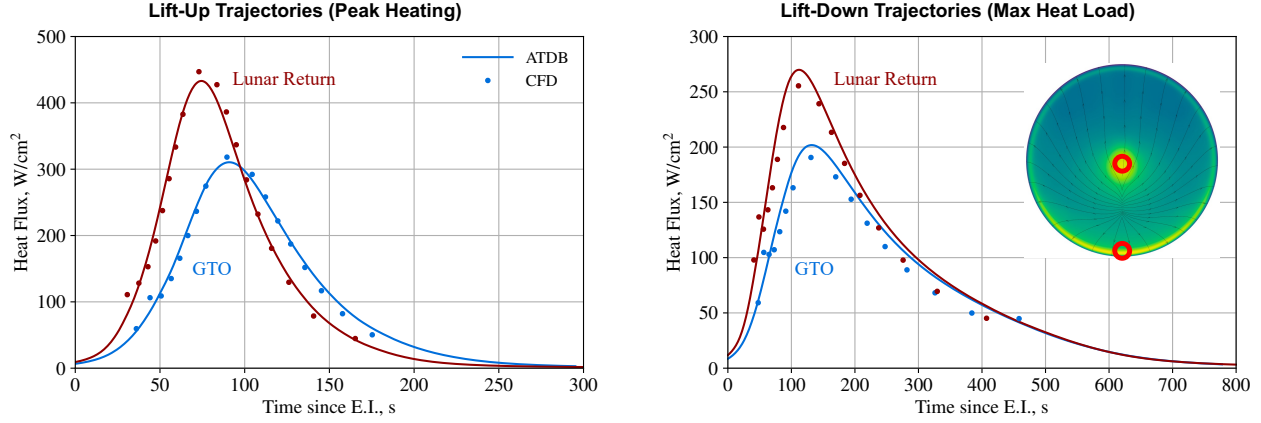


Fig. 6 Predicted convective heating versus time at the body points associated with the peak heat flux for each bounding trajectory. Body points are identified in right-most figure inset.

agreement between the ATDB and the validation CFD is excellent. With the fastest entry speed, the lunar return trajectory has the largest convective heat flux, peaking around 430 W/cm^2 for the lift-up trajectory. The peak heat flux expected for the GTO lift-up trajectory is 310 W/cm^2 . Both lift-down trajectories experience significantly reduced peak heating, as shown in the figure. However, the integrated heat loads are higher for the lift-down cases, due to the increased length of time spent in the aerocapture maneuver. The largest heat load predicted was obtained for the lift-down lunar return trajectory, at 66 kJ/cm^2 .

Similar figures are presented for the pressure distributions in Fig. 7. Again, excellent agreement is obtained between the ATDB and validation dataset. The peak wall pressure is expected during the lift-up lunar return trajectory, coming in just shy of 20 kPa . As with heating, the instantaneous pressure environment for the lift-down trajectories is substantially decreased compared with the lift-up environments. A similar picture emerges for the radiative heating and shear stress distributions but are not shown here for brevity. However, the so-called butterfly plots of Fig. 8 show a more complete picture of the estimated instantaneous aeroheating environments for each reference trajectory. For each reference trajectory, the ATDB was used to estimate time-dependent aeroheating quantities of interest at all body points along the vehicle forebody centerline and a convex hull is placed around the resulting point cloud in shear-pressure and heating-pressure space. This analysis provides a more complete picture of the forebody aeroheating environments that the TPS must withstand. From the figure, it is clear that the most extreme environments are predicted for the lift-up trajectories. This is expected since the vehicle reaches a lower altitude in the lift-up configuration (Fig. 4) and experiences peak freestream densities that are nearly double those seen in the lift-down trajectories. For reference, the centerline CFD solution from the lunar return peak heating condition is also plotted in Fig. 8. This shows the instantaneous environment on the centerline at that trajectory condition and helps explain the shapes of the convex hulls. As can be seen in the figure, the low pressure side of the peak shear stress edge corresponds to the windside shoulder of the vehicle, while the high pressure side can be attributed to the stagnation region. The stagnation point falls on the zero shear stress axis, and accounts for the largest pressure, as expected. Likewise, we see two peaks in the heat flux along the centerline, one at the windside shoulder due to the compression of the boundary layer as the flow wraps around the sharp edge of the vehicle, and the other at the nose.

A complete summary of the peak aeroheating environment estimates is provided in Table 2, including values for a reference Uranus aerocapture mission [6] for comparison. From an aeroheating perspective, the most stressing environments are found to be associated with the lunar return trajectory. However, from a technology demonstration perspective, the GTO option appears to be more similar to the reference Uranus aerocapture mission, than a lunar return trajectory. One aspect of the environment that cannot be reproduced with either trajectory option is the total heat load experienced by the vehicle. Due to the substantially larger atmosphere of Uranus, an aerocapture vehicle will spend significantly more time in the aerocapture maneuver, leading to a heat load of 83 kJ/cm^2 for the reference mission. At Earth, the two trajectory options in this study were only able to provide up to 66 kJ/cm^2 . However, this is less of a concern than the peak heating rates, because the heat load typically drives the TPS thickness, and not the choice of material. More discussion of the impact of the estimated aeroheating environments are presented in Sec. V.

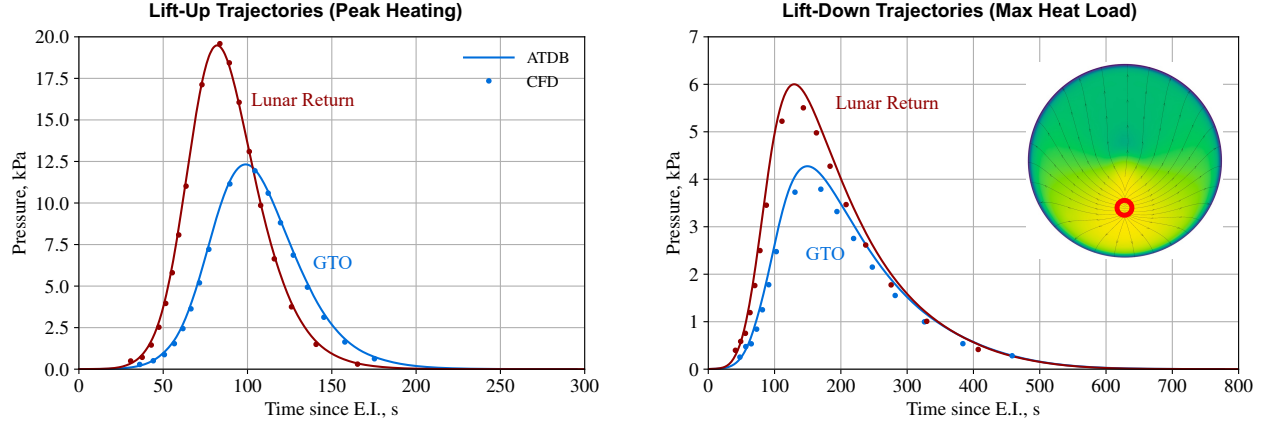


Fig. 7 Predicted pressure versus time at the body points associated with the peak pressure for each bounding trajectory. Body points are identified in top-left figure inset.

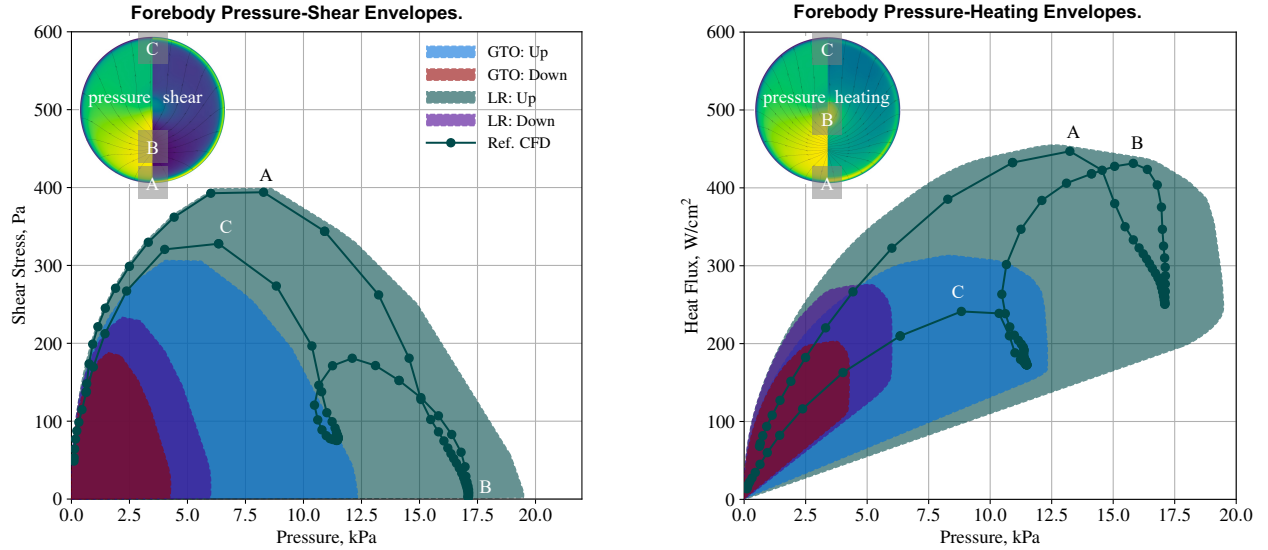


Fig. 8 Butterfly plots showing the instantaneous aeroheating environment space on the forebody of the vehicle for each bounding trajectory. A reference CFD solution near the peak heating condition for the lift-up lunar return trajectory is also shown, with points running along the forebody centerline. Points A, B, and C are identified to orientate the curve on the vehicle geometry.

E. Targeted Aeroheating Studies

This subsection presents results of targeted aeroheating analyses which were used to inform various vehicle design decisions early in the design process.

Table 2 Summary of peak aerothermal loads for the ARRIVAL trajectory concepts.

Scenario	Conv. Heating W/cm ²	Rad. Heating W/cm ²	Total Heating W/cm ²	Heat Load kJ/cm ²	Pressure kPa	Shear Stress Pa
Lunar Return	433.0	44.0	455.0	66.0	19.5	398.7
GTO	310.4	8.4	313.2	54.3	12.3	306.4
Uranus Reference	350.0	-	350.0	83.0	8.0	130.0

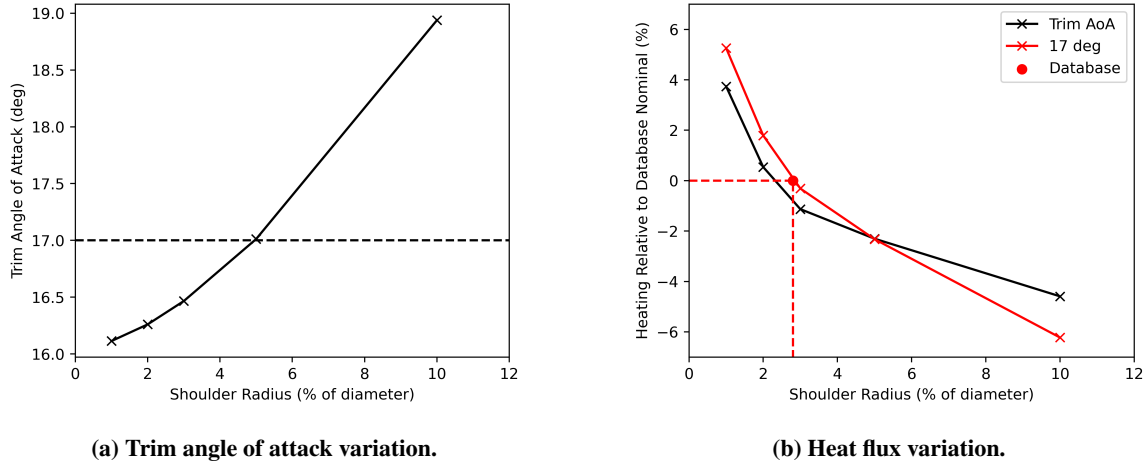


Fig. 9 Variation of trim angle of attack (a) and peak surface heat flux (b) with shoulder radius (percent of vehicle diameter).

1. Vehicle shoulder radius

During the design and layout of the vehicle's interior cavity and payload, changes to the vehicle's outer mold line (OML) were considered. To provide context for the impact of changes to the vehicle's OML on the aerothermal environment, a study was performed to assess the sensitivity of the peak convective heat flux to changes with respect to the vehicle shoulder radius. For this study, the most severe condition in the potential trajectory space shown in Fig. 4 was considered. A range of shoulder radii from 1 to 10 percent of the vehicle diameter was simulated, bounding the nominal value of 2.8%. As changes in shoulder radius affect the lift to drag ratio of the vehicle, the pitch angle which achieves the nominal ratio of 0.25 at each shoulder radius was determined first. The trim angle for the vehicle as a function of shoulder radius is shown in Fig. 9a. The peak heating on the vehicle shoulder as a function of shoulder radius is shown in Fig. 9b for both the trim and nominal angles of attack. The peak heating rises with decreasing shoulder radius as expected, but factoring in the varying trim angle attenuates the variation slightly. This is most noticeable at the highest shoulder radius, where the reduction in heating compared to the database nominal value is attenuated by approximately one third. Ultimately, when considering the large change in trim angle of attack incurred by increasing the shoulder radius, reductions in heating are minimal and not likely to benefit TPS design.

2. Aftbody heating and impact of turbulence

To ensure that the assumed aftbody heating environments used in the present study were conservative, a solution of the vehicle wake was performed for the peak heating conditions of both the lift-up and lift-down trajectories. As both are similar, only the lift-up trajectory results are presented here. In the present study, the aftbody heating environments used for TPS sizing are assumed to be 30% of the forebody heating result from the ATDB. To assess the conservatism of this assumption, the heating on the vehicle aftbody was predicted with LAURA using a full vehicle wake domain. Additionally, the sensitivity of the heating to boundary layer turbulence was assessed by comparing the laminar solution with various turbulence models. Figure 10 shows the surface heat flux on the vehicle centerline for both the forebody and aftbody. Immediately it is clear that for this condition, the aftbody heating assumption for TPS sizing is conservative compared to predictions, even considering a margin on the order of 100%. Additionally, the results demonstrate a relatively weak heating augmentation due to turbulence. On the forebody, the maximum turbulent heating augmentation occurs downstream of the stagnation point along the leeside of the forebody, but does not exceed the maximum laminar heat flux on the forebody. On the aftbody, similarly low augmentation is observed at the present conditions. While detached hypersonic wakes are challenging to simulate accurately [19], the model form error in these predictions is not expected to exceed the assumptions used in the TPS sizing analysis.

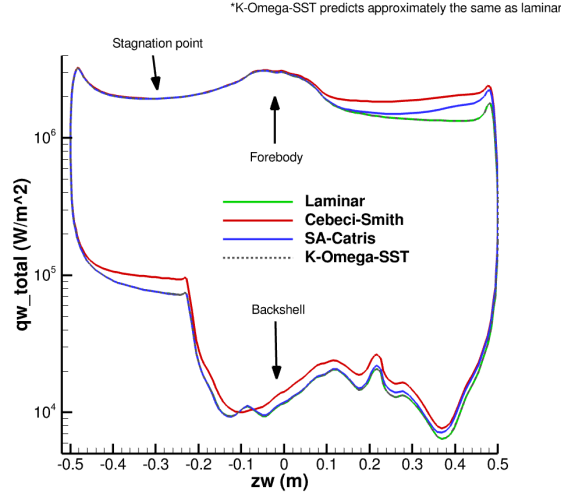


Fig. 10 Surface heat flux along the vehicle centerline including the aftbody.

3. Parametric uncertainty analysis

In order to estimate the uncertainty in the heating predictions due to uncertainties in the chemical reaction rate and diffusion models, a parametric uncertainty quantification analysis was performed. A total of 41 input variables which determine the chemical reaction rate models and diffusion coefficients were considered and treated as epistemic uncertainties with no known distribution. For the present analysis LAURA UQ, which is a module within the LAURA CFD code, was utilized [18]. LAURA UQ uses a non-intrusive polynomial chaos (NIPC) expansion to build a surrogate model that can be used to propagate input uncertainties through the CFD model efficiently [20]. Typically, the number of high fidelity samples needed for an NIPC sample is equal to the number of terms in the polynomial chaos matrix [20] which for a 2nd order expansion and 41 variables would be 903. LAURA UQ uses a least angle regression procedure to undersample the matrix and achieve an accurate model with minimal samples required [18]. A total of 50 CFD solutions were used to propagate the input uncertainties for this model, and a second analysis with 100 samples was used to verify that the analysis was sufficiently converged. Figure 11 shows the parametric uncertainty bounds for the heat flux on the vehicle forebody along the centerline at GTO lift-up peak heating conditions. The results show that parametric uncertainties due to the thermochemical model account for approximately $\pm 5\%$ on the vehicle forebody for this condition, which is bounded by the assumed margins for the present study. It is important to note that this analysis is not assumed to be representative of all database conditions or the vehicle aftbody where turbulence model form uncertainty is expected to dominate. Instead, this result provides confidence that thermochemical modeling uncertainties are within the margins assumed for the present study.

V. TPS Analysis

Understanding the aerothermal environments experienced by the aeroshell allows for the TPS design work to begin. Starting with a brief discussion of the TPS sizing methodology and assumptions, this section presents the selection of TPS accompanying sizing of the chosen material to meet the thermal demands. The scheme for sizing the thermal protection system is explained, and the sub-selection of body points for the vehicle is presented with accompanying predictions for TPS thickness across the body points for the lift-up and lift-down flight profile within the GTO and LRO con-ops. Next, a discussion of risk-reduction activities to accomplish in ground-testing is covered, as these trajectories are among the highest heat loads ever predicted for a NASA mission. Finally, a brief substructure sensitivity study was completed to inform the project investigation into a composite structure instead of the baselined titanium structure, as a primary vehicle integrator has not been selected for ARRIVAL at this early stage.

A. TPS Sizing Methodology and Assumptions

TPS typically consists of the following components: a material system that is highly efficient at heat rejection usually installed as tiled pieces, a compliant adhesive to fill the gaps between tiles, and another adhesive to bond

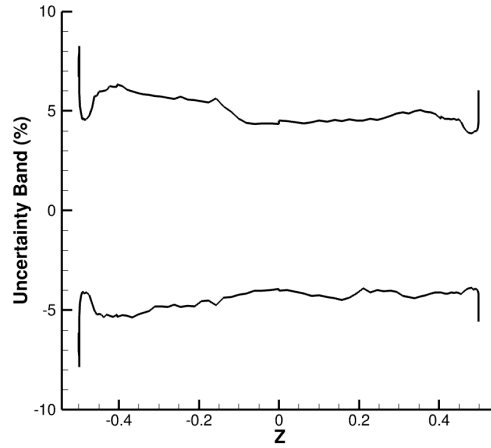


Fig. 11 Parametric uncertainty bounds for the vehicle forebody centerline heating at GTO lift-up peak heating conditions.

to the vehicle substructure. The selection and design of specific materials is heavily influenced by the aeroheating environments discussed previously in Section IV. The chosen material for ARRIVAL is similar to work done for a Uranus aerocapture mission study albeit the substructure is different[7]. Figure 12 shows the materials used in the stack-up and the thicknesses modeled. Conformal Phenolic Impregnated Carbon Ablator (C-PICA) is a low-density carbon felt that is infused and cured with phenolic resin. During high-heating, the phenolic pyrolyzes and blows decomposing material into the boundary, inhibiting heat transfer to the spacecraft. C-PICA is in the family of low-density ablators offered by NASA and is similar to PICA used for MSL and Mars2020, and Domestic PICA (PICA-D) which is the heatshield material for the Dragonfly mission to Titan. C-PICA has been flown before several times aboard the Varda Space Industrie’s Winnebago vehicle successfully, and has been tested in the NASA Ames arc-jets using air as a test gas to heat fluxes in excess of 700 W/cm^2 as a tiled test article without failure. Owing to the size of the vehicle, thin RTV-560 gap-filler is used between C-PICA tiles to prevent heated flow from reaching the bondline. The C-PICA material is bonded to a project-baselined titanium structure using RTV-560. Due to coefficient of thermal expansion disparities between the titanium and C-PICA, a strain isolation pad (SIP) is necessary to prevent mechanical failure of the system when heated. The SIP itself has flight heritage from use aboard the Space Transport System (STS) shuttle. While the substructure is sized based on vehicle structural requirements, the gap-filler and RTV-560 thicknesses are selected for manufacturing feasibility and sufficient bond strength. The outermost layer, C-PICA, is sized to achieve a target bondline adhesive temperature at a given body point along a specific trajectory, having been selected based on demonstrated performance in arc-jets and previous Earth missions [21, 22]. C-PICA was also baselined as the forebody TPS for the Uranian reference mission in previous work[7].

Because sizing at a single body point on a single trajectory does not capture all system risks and uncertainties across the vehicle, a systematic approach is used to account for material property variability, aerothermal environment uncertainties, and trajectory dispersions that may alter the entry profile. The process begins with selecting trajectories expected to generated the highest heating rate and heat loads derived from aerothermal indicators based on CFD analyses presented in Section IV. This approach accounts for trajectory dispersions and will be referred to as the nominal trajectories for the purposes of the TPS sizing methodology.

With trajectories selected, the NASA Fully Implicit Ablation and Thermal (FIAT) tool is used to perform one-dimension thermal analysis and determine the required TPS thickness [23]. FIAT iteratively calculates the temperature at a user-defined interface—here, the bondline adhesive, which is critical for TPS attachment. The bondline temperature constraint is set to the adhesive’s single-use limit provided by the manufacturer unless otherwise stated.

To ensure robustness against known and perceived risks, a three-branch sizing approach is used[24]:

- 1) Nominal branch: no uncertainty modifiers except for the trajectory uncertainty capture
- 2) Aeroheating branch: includes augmented heat flux and pressure to account for aerothermal uncertainties
- 3) Thermal branch: incorporates uncertainties in material modeling and properties

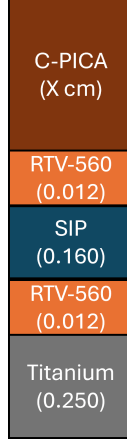


Fig. 12 ARRIVAL Forebody Material Stack-up, thickness shown in inches.

While the aeroheating branch makes use of multipliers to the heat transfer coefficient and applied pressure, the thermal branch instead modifies the acceptable bondline temperature. A Monte Carlo simulation (mcFIAT) randomly samples key material properties informed through provided Gaussian distributions. The difference between the 2σ peak bondline temperature and the average bondline temperature from the mcFIAT analysis is used to reduce the acceptable performance of the TPS, thus applying a material related margin. The final TPS thickness is computed by combining the three branches using the root-sum-square (RSS) method:

$$t_{\text{RSS}} = t_{\text{nominal}} + \sqrt{(t_{\text{aeroheating}} - t_{\text{nominal}})^2 + (t_{\text{material}} - t_{\text{nominal}})^2} \quad (3)$$

This yields a thickness that is robust to environmental and material uncertainties. However, additional allowances may be required to account for project-specific factors and manufacturing tolerances appropriately added depending on mission risk posture. The sizing presented in the results section does not add any mission factors or tolerances to the TPS thickness.

B. TPS Sizing Results

1. Analysis Body Point Selection

Knowing the materials involved, the sizing methodology applied, and the analysis tool from the previous subsection, all that is required is selected body points on the vehicle to perform the sizing analysis. The one dimension nature of FIAT leads to design work focused on a handful of discrete locations chosen to sample the re-entry vehicle forebody as opposed to examining the entire body like the CFD analysis. Taking a bisection of the vehicle in Fig. 13, four body points were taken along both the windward and leeward side of the vehicle with an additional point at the vehicle nose. The windward body points were meant to roughly evenly distribute while honing in on locations of interest. Body point 61 is the vehicle stagnation point while body point 82 was selected to investigate higher heating just upstream of the shoulder geometry. The leeward side is a mirror of the windward.

2. Environment Sampling

Sampling the ATDB using the collection of body points for the lunar return Lift-Up trajectory provides location specific heating rate environments that can be used to support TPS sizing, seen on the left in Fig. 14. The integrated heat load for each body point is calculated and shown in the legend. Peak heating occurs at the shoulder but the nose point tracks closely and actually experiences higher heat loads due to slightly higher heating at the tails of the heating pulse. A better cross comparison of the GTO and lunar return environments for the Lift-Up and Lift-Down sizing cases can be viewed from the right side plot in the same figure.

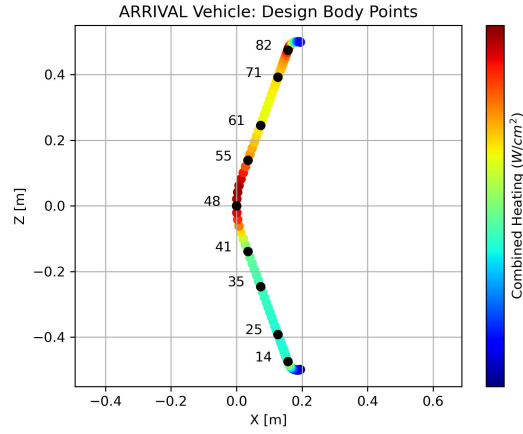


Fig. 13 ARRIVAL Selected Forebody TPS Design Points.

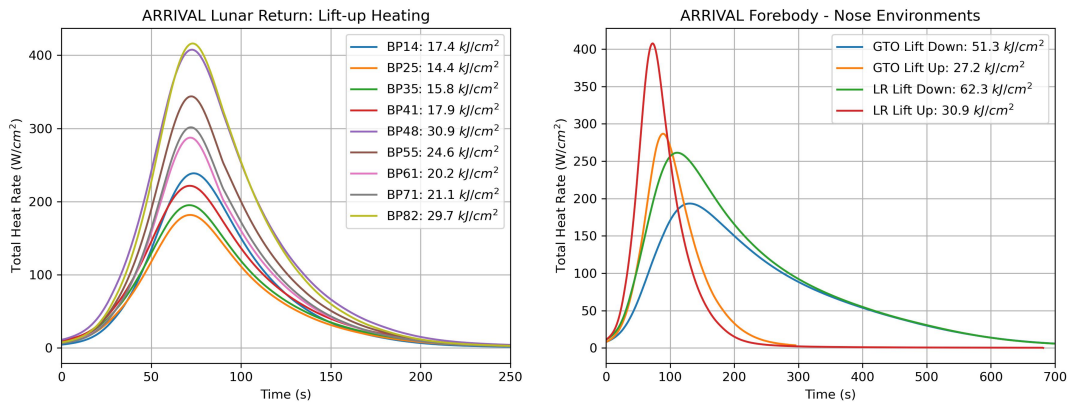


Fig. 14 ARRIVAL Lift-Up Environment Profiles at Selected Body Points.

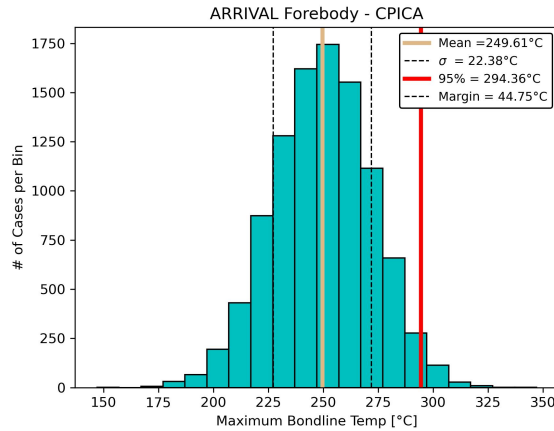


Fig. 15 MCFIAT Peak Temperature Histogram for the Design-Driving Case.

3. Forebody TPS Sizing

As previously discussed, TPS sizing was performed using the RSS methodology for each of the trajectories at body-specific locations. The aerothermal uncertainty was accounted for by using simplified 35 percent increase in the heat transfer coefficient and 10 percent in pressure. These margins are expected to be bounding of the parameter uncertainty discussed in Section IV, however forward work is required to verify this.

MCFIAT analysis incorporated Gaussian 2σ uncertainty material properties which were adopted from the Mars2020 project which used an older material version, PICA-D. The analysis sampled properties to create 10,000 unique FIAT simulations and tracked peak bondline temperatures in nominal, unmarginated runs. The 95 percent bounding peak bondline temperature was compared to the average peak bondline temperature to create a thermal margin to use in derating allowable performance in the third branch of sizing. Peak bondline temperature, a signal of how much thermal energy is getting through the TPS, is most sensitive to virgin and char conductivity but uncertainties for each material and a wide range of properties were incorporated. The uncertainties used for ARRIVAL MCFIAT simulations are the same as those used in the reference mission study[7]. The vehicle structure for ARRIVAL is simplified compared to that mission study and therefore not all of the uncertainties were used. For ARRIVAL, the facesheet uncertainties were applied to the titanium.

After completing nominal branch screenings to identify the likely driving body point and trajectory pairing, a MCFIAT simulation was completed for the lunar return lift-down trajectory at the nose, BP48. The histogram in Fig. 15 shows binned peak bondline temperatures, the 95-percentile peak temperature, and the resultant thermal margin. This thermal margin was conservatively used for all body point locations and trajectory analyses presented below to simplify analysis efforts during this preliminary phase of the mission.

Completing the RSS sizing process gives the C-PICA thickness results in Table 3. The heat fluxes are well within the capability of the thermal protection system and thus the performance is driven by the heat load. This is reflected in the final thickness solutions with the lift-down requiring more insulation, approximately 50 percent more material compared to lift-up solutions. The side of the corridor the vehicle flies along is by far the most influential parameter. Outside the shoulder region, the windside of the vehicle requires only about 5 percent more C-PICA thickness over the leeside. Similarly, the GTO and lunar return mission architectures do not significantly impact the TPS thickness solution, although an increase is generally seen when choosing lunar return. The design-driving body point is the vehicle nose which would need 4.8 cm and 5.4 cm for the GTO and lunar return missions, respectively. From a TPS perspective, the differences between the two options is small. The approximately 100 W/cm^2 difference in peak heating would not significantly impact ground test programs. The surface temperatures observed in Figure 16 for unmarginated thickness solutions do not alter the oxidation regime of ablation, and the required thicknesses are of a similar effort in manufacturing for both C-PICA and the gap filler. The mass differences between the mission architectures do not impact the vehicle overall and ultimately a C-PICA system is appropriate for both. Limited mass savings can be gained from tailoring the TPS thickness across the vehicle. However a manufacturing simplification is preferred, setting the entire heatshield thickness to the maximum value found. For all trajectories, this would be at the nose.

An estimate of the expected C-PICA response from the bounding analysis case is shown in Figure 17. A few key

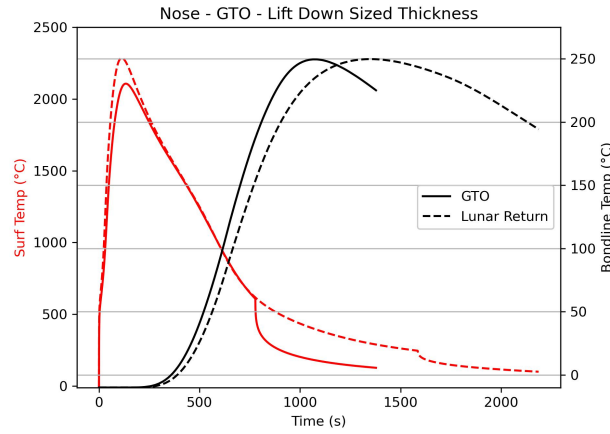


Fig. 16 Unmargined C-PICA Surface and Bondline Temperature Histories with 10 minutes of Simulated Cooldown.

observations are made. Over the first 500 s of flight, C-PICA's surface will recess a little over 1 cm in Earth's oxidizing atmosphere. Second, C-PICA's margined thickness results in the design bondline temperature of 250 °C getting only about 80 % of the way though the TPS. Lastly, the peak expected bondline temperature is less than 200 °C. C-PICA decomposition estimates the TPS material fully charring to around 2 cm with the deepest approximately 1 cm still fully virgin. Regarding recession, the vehicle flank experiences around 0.5 cm recession while the nose for a GTO lift-up trajectory is about 0.85 cm. This ablation response is both similar to and different from the Uranus study completed in [7]. The oxygen in Earth's atmosphere allows for oxidation processes to remove energy through recession whereas in the Uranus study, material was only lost when surface temperatures reached sublimation temperatures. The char depth is relatively similar between ARRIVAL and the Uranus study.

4. Test Limits and Necessary TPS Test Program

Flight experience is considered the gold standard for demonstrating performance as no ground-test facility or model perfectly captures the highly complex plasma environment. Even so, the current experience set of C-PICA in Varda Space Industries earth return trajectories is not the same as those predicted by ARRIVAL. In addition, it is very difficult to capture high quality, comprehensive data aboard spacecraft as mass and data transfer incur significant costs to the program. Testing in arc-jets allows material scientists and programs to simulate the environment in advance and interrogate the material with an assortment of instrumentation to add fidelity to the models and mitigate failure modes and risk to the program before flight. C-PICA has been tested in the NASA Ames arc-jets using air as a test gas up to

Table 3 Final C-PICA sized thickness results, in cm.

BP	GTO		Lunar Return	
	Down	Up	Down	Up
14	4.0	2.6	4.2	2.6
25	4.0	2.6	4.2	2.7
35	4.0	2.7	4.4	2.8
41	4.1	2.7	4.5	2.9
48	4.8	3.1	5.4	3.3
55	4.3	2.8	4.6	3.0
61	4.1	2.8	4.4	2.8
71	4.2	2.8	4.4	2.8
82	4.5	2.9	4.8	3.0

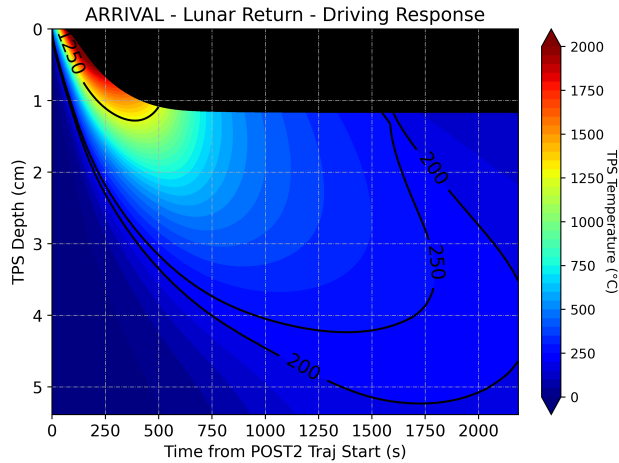


Fig. 17 Margined thickness C-PICA temperature contour for driving design analysis.

heat fluxes of 700 W/cm^2 . The heat flux demonstrates that the material can handle high-heating, but aerocapture does not incur a massive heat flux – instead it is the heat load that is significant. To sufficiently retire risk for an aerocapture demonstration mission, the program would instead focus on the performance of the material under long exposure to the plasma environment at moderate conditions. The long exposure will stress both the insulation capacity of the material and the estimations of the material performance that accompany it. During a long entry, there is ample time for dissociated oxygen to interact with the material, decaying the substrate carbon felt and residual char. This will lead to higher recession on the surface as well as high-porosity within the material. In the same vein, the gap-filler will be exposed to a similar environment. The dissimilarity in recession between the gap-filler and TPS may have unintended consequences for the recession and heat flux distribution around the vehicle, and could lead to roughness and turbulent augmentation not captured by idealized CFD this early pre-phase A stage of study. Finally, for C-PICA, because of the amount of energy ingested by the material, conductivity will become even more critical to understand as this may have broad implications for the efficiency of the system under such a long dwell. Indeed, multi-dimensional analysis may need to be leveraged with appropriate data from a ground test in order to adequately assess the performance of C-PICA during the aerocapture maneuver. These factors will most heavily influence the uncertainties quantified in the thermal margin branch and could lead to higher overall thickness for the final TPS sizing in later stages of the ARRIVAL project.

The remaining risk mitigation activities that a test program may address focus on the scale of the TPS implemented over the size of the vehicle targeted. RTV gap-filler for Mars2020 and Orion vehicles is designed to be a minimal in weight and exposure to outside elements, but is balanced by the practicalities of machining and installing an array of tiles over the acreage of the vehicle aeroshell. Given the thickness and compliance of C-PICA, the gap-filler for the heatshield may need to be implemented differently and may yet be modified entirely when testing is formally begun. In addition, the mechanical properties of the TPS at scale, and as an integrated system with the aeroshell substructure, may reveal knockdowns in performance that are attributed to the thickness of the C-PICA, since it has never before been used in this way.

5. Aftbody TPS Sizing

Aerothermodynamic work was completed to provide an expected bounding heating rate for the aftbody. Despite the limited aftbody information a TPS material screening was performed to understand viable options. For sizing, the forebody stagnation point (BP61) was scaled to 30 percent of the heat transfer coefficient and 10 percent of the pressure; this is considered a very conservative sizing. The strain isolation pad was removed for the aftbody and an extra 20 percent thickness margin was added to cover the additional uncertainty due to not having anchoring CFD.

Several material choices were analyzed, each having different attributed. C-PICA would simplify the procurement process for a flight mission by matching the forebody TPS. Lockheed Martin's Super Lightweight Ablator (SLA-561V) is the most historied system while Acusil-II by Peraton offers RF-transparency. Finally Materials Engineered for Re-entry using Innovative Needling Operations (MERINO) is a new NASA TPS product that may reduce price and schedule to implement. Because MERINO is a new material, an extra 15 percent thickness margin is added to account

for changes as the material and its thermal model matures. Table 4 shows the sized thickness of each candidate material for each design trajectory. In all cases, the material thicknesses are manufacturable. In general lunar return missions would require about 10 % more thickness. However, material density differences must be included when evaluating across material systems. Combining the thickness and density of the material creates an areal density value of unit kg/m^2 , which indicates the mass efficiency of the system. MERINO is estimated to be the most mass efficient with an areal density of $2.55 kg/m^2$, next comes SLA-561V, then C-PICA at $6.5 kg/m^2$, and finally Acusil-II is the least efficient to gain the RF-transparency benefit. Each of these material would work under ARRIVAL's conservative aftbody environments. The choice would likely be driven by other project and procurement constraints.

Table 4 Final aftbody sized thickness results, in cm.

Material	GTO		Lunar Return	
	Down	Up	Down	Up
C-PICA	2.20	1.07	2.38	1.06
SLA-561V	1.12	0.54	1.22	0.59
Acusil-II	3.78	1.74	4.10	1.70
MERINO-LD	1.96	0.94	2.13	0.94
MERINO-LD (+15%)	2.26	1.08	2.45	1.08

C. Substructures Sensitivity Trades

Toward the close of the pre-phase A study, ARRIVAL revisited the structure material choice of titanium and was investigating the impacts across various subsystems if a composite substructure was used. This change offers several benefits to the thermal protection system all center around the SIP. Firstly, the titanium structure required a SIP due to CTE mismatch but a SIP acts as a thermal insulator too which prevents the TPS from using the structure as a heat sink. Switching to a composite eliminates the SIP which in turn saves about 0.4 cm in thickness. About $0.5 kg/m^2$ is also removed from the total system areal density. Removing the SIP also removes one of the RTV bonds which saves approximately another $0.36 kg/m^2$. Finally, because the thermally isolating SIP is not present, C-PICA can heat sink to the composite structure thus decreasing the thickness and mass of the TPS. For the GTO solution, TPS thickness reduces from 4.8 cm to 4.25 cm, roughly a $1.5 kg/m^2$ savings. In total, about 1 cm of the TPS system is removed, which is doubled when evaluating diameter constraints. This thickness reduction does not account for thickness changes of the structure which also reduced. The total areal density reduction is about $2.4 kg/m^2$.

VI. Conclusions

An Earth aerocapture technology demonstration has been presented through the lens of aeroheating and necessary thermal protection system. The concept of operations for two different trajectories was shown, GTO and lunar return, both starting from different initial conditions and yielding different entry profiles that lead to the same exit condition in order to target the same post-capture orbit. The aerothermal database was introduced, allowing the analysts to more easily interrogate the entry space with a validated surrogate model that outputs aeroshell conditions to be used by downstream analysis pipelines. The highest heating environment originates from the Lunar Return trajectory as expected, while the GTO is more similar to the Uranus reference mission studied by an earlier NASA project. To withstand the predicted environment, TPS was sized using a three-branch RSS approach to account for uncertainties, and it was found that the NASA C-PICA material could be adequately manufactured to meet the aerothermal needs. Data gathered and lessons learned from manufacturing would feed directly into a Uranian aerocapture mission as previous studies baselined C-PICA. Aftbody TPS needs were calculated by scaling the forebody environments as is common with pre-phase A studies, and it was shown that traditional TPS as well as novel NASA TPS stacks are capable for implementation, with the opportunity to leverage experience to future Uranian aerocapture missions. Comments on gaps to address potential risks in TPS implementation were made, and will mature as work progresses.

Acknowledgments

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